horizon's evolution to be teleological. Since teleological viewpoints have rarely if ever been useful in modern physics, the absolute horizon seemed hardly worth exploring.

Box 12.2

Evolution of an Accreting Hole's Apparent and Absolute Horizons

The spacetime diagram below illustrates the jerky evolution of the apparent horizon and the teleological evolution of the absolute horizon. At some initial moment of time (on a horizontal slice near the bottom of the diagram), an old, nonspinning black hole is surrounded by a thin, spherical shell of matter. The shell is like the rubber of a balloon, and the hole is like a pit at the balloon's center. The hole's gravity pulls on the shell (the balloon's rubber), forcing it to shrink and ultimately be swallowed by the hole (the pit). The *apparent horizon* (the outermost location at which outgoing light rays—shown dotted—are being pulled inward) jumps outward suddenly, and discontinuously, at the moment when the shrinking shell reaches the location of the final hole's critical circumference. The *absolute horizon* (the boundary between events that can and cannot send outgoing light rays to the distant Universe) starts to expand *before* the hole swallows the shell. It expands *in anticipation* of swallowing, and then, just as the hole swallows, it comes to rest at the same location as the jumping apparent horizon.